

GANESH RAJ KYATHAM

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SUMMARY

AI Engineer with 4.5 years of experience building scalable GenAI solutions, agentic RAG systems, and MLOps pipelines. Specialized in end-to-end LLM orchestration, tool calling, and automated evaluation frameworks to ensure backend reliability. Proven track record of shipping high-throughput conversational assistants, collaborating closely with cross-functional stakeholders to deliver resilient, enterprise-grade AI systems.

EXPERIENCE

AI Engineer , Fedway Associates - EliteUS	<i>Jan 2025 - Present, Edison, NJ</i>
- Built and shipped LLM-powered conversational AI assistants for B2B trade and compliance queries using Python, FastAPI, AWS Bedrock, and LangGraph, automating customer support and handling 20,000 requests per week across registered accounts. - Implemented RAG combining hybrid retrieval (BM25 + dense vector + RRF) and structure-aware chunking with parent-document mapping to execute compliance queries with grounded generation. Integrated human-in-the-loop (HITL) workflows, prompt engineering, and RAGAS evaluation frameworks with golden datasets, achieving a 60% support deflection rate and 85% Recall@10 using hybrid retrieval. - Built CI/CD and MLOps pipelines with automated regression testing using GitHub Actions, Docker, and AWS ECR. Implemented production monitoring, distributed tracing, Guardrails, LLM observability, and custom evaluation metrics using LangSmith and AWS CloudWatch to maximize system reliability. - Owned end-to-end delivery of production AI backend services from prototyping through deployment, leading model selection, A/B testing, and performance optimization. Collaborated with cross-functional supply chain stakeholders and mentored interns throughout the development lifecycle.	
Data Scientist - NLP , UCM Rutgers	<i>Feb 2023 - Dec 2024, Remote</i>
- Built an NLP pipeline by fine-tuning DistilBERT models in PyTorch, utilizing MLflow for experiment tracking and model registration to automatically classify 300+ daily university blogs into 600+ categories across 33 publishing sources. - Developed automated inference, indexing, model serving, and deployment workflows using Docker, GitHub Actions, and Azure, increasing cross-pollination CTR by 35% while eliminating zero-result searches.	
Data Scientist - Applied AI , Deloitte Consulting USI	<i>May 2019 - Jan 2022, India</i>
Exelon: Developed object detection models using yolo and PyTorch on 4.5TB of satellite imagery, reducing audit cycles by four months while improving utility asset data completeness. - Developed geospatial computer vision datasets and processing pipelines using Python, Rasterio, GDAL, and PostGIS, performing image preprocessing, annotation, geolocation, and feature extraction across 1,200 zip codes. NGA: Developed computer vision models for automated dock monitoring across three ports, enabling vessel and infrastructure analysis that reduced revenue leakage by 15% while eliminating over 150 hours of manual effort each week.	
Data Engineer , MAQ Software	<i>May 2018 - July 2018, India</i>
Migrated legacy SSIS ETL workloads to Azure Data Factory (ADF), integrated three data marts into Azure SQL Database; developed robust SQL stored procedures for data validation, deduplication, and database integrity.	

EDUCATION

M.S. in Data Science , Rutgers University - New Brunswick	<i>Aug 2022 - May 2024</i>
B.Tech. in Computer Science & Engineering , Indian Institute of Technology, Indore	<i>July 2015 - May 2019</i>

CERTIFICATIONS & SKILLS

- AWS Certified Machine Learning Specialist
- AWS Certified Cloud Practitioner

Programming **Generative AI**: Python, Generative AI, LLMs, Retrieval-Augmented Generation (RAG), AI Agents / Agentic AI, Agent Orchestration, Tool / Function Calling, Structured Outputs, Prompt Engineering, Embeddings, Semantic Search, Hybrid Search (BM25), Reranking, Chunking, Vector Databases (Pinecone, ChromaDB), LangChain, LangGraph, AWS Bedrock, OpenAI API, Claude, Hugging Face

Machine Learning: PyTorch, Transformers, Fine-tuning, Transfer Learning, Scikit-learn, spaCy, NLP, Computer Vision, Classification, Model Evaluation, Evaluation Frameworks, Hallucination Detection

Backend MLOps: FastAPI, REST APIs, Microservices, Docker, GitHub Actions, CI/CD, MLOps, Model Serving, LangSmith, LLM Observability, SDLC, MLflow, AKS

Cloud Data: AWS (Bedrock, Lambda, ECR, SQS, CloudWatch, ALB), Azure OpenAI, Databricks, PostgreSQL, MongoDB, SQL

PROJECTS

- EasyApply Bot**: [blogpost]Built a job application automation system using Python, Playwright as MCP, Selenium, and hybrid RAG for answering questions. Processed 200 job applications under 4 hrs.
- 3D Space Scanner**: [project] Designed and implemented a photo based 3D dimension checker using Apple’s multimodal DepthPro, OneFormer, HuggingFace and LSUN, achieving 85% accuracy.